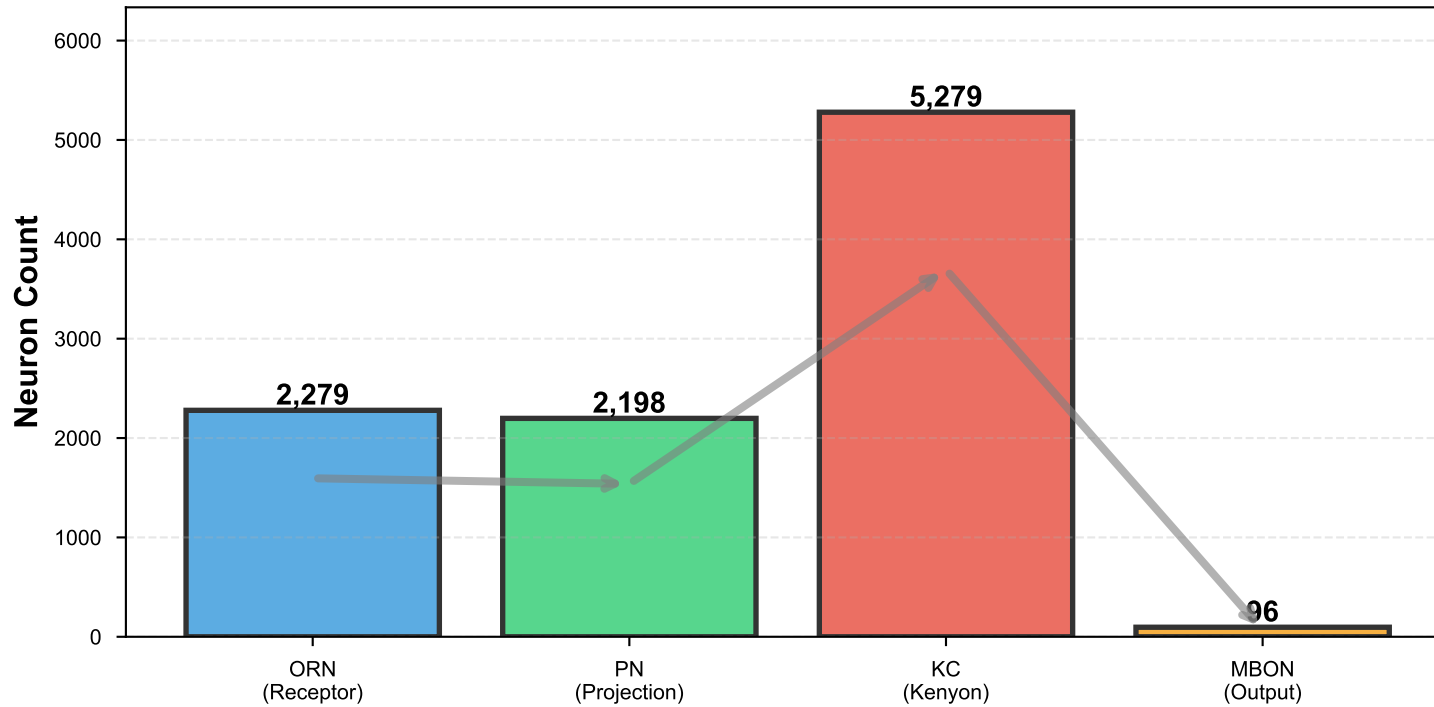


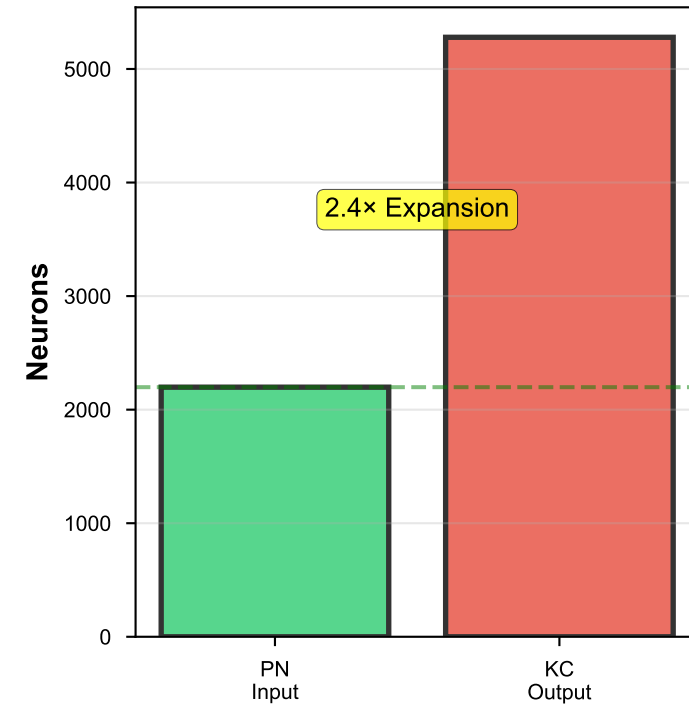
Connectome Architecture: Wave-Based Full Brain Simulation

Olfactory Pathway Structure and Connectivity

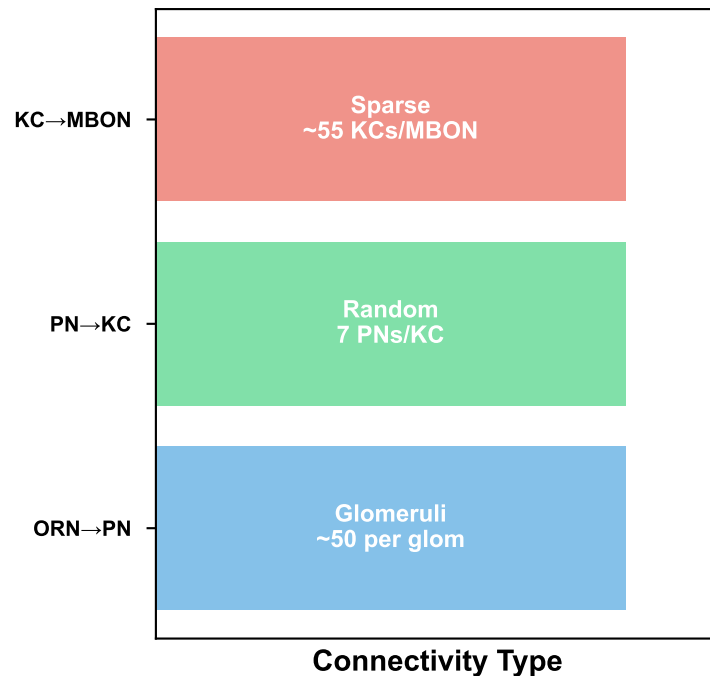
A. Olfactory Pathway Architecture
Complete Connectome (139,255 total neurons)



B. Sparse Expansion
2.4× Dimensionality



C. Connection Patterns



NETWORK STATISTICS

Olfactory Pathway:

- Total neurons: 10,906 (7.8% of brain)
- Total synapses: 446,388
- Average connectivity: ~41 synapses/neuron
- Sparse expansion ratio: 2.4× (PN→KC)

Key Properties:

- Random PN→KC wiring (Caron et al. 2013)
- Each KC samples ~7 PNs
- High activation threshold (5+ inputs needed)
- Global inhibition via APL neuron

Full Brain Context:

- 139,255 total neurons
- 5,342,446 total synapses
- Memory: 64 MB (0.46 bytes/neuron)
- Simulation: 10× real-time